

Project Definition

BACKGROUND

Existing literature highlights the unique challenges faced by neurodivergent individuals, particularly in areas such as mood regulation, symptom management, and executive functioning. Research shows that commonly used and available tools often fall short in providing personalized support and actionable insights for this demographic. Studies also emphasize the importance of user-centred design and customizability in applications for neurodivergent users, as individual needs and experiences vary widely. Despite advances in mental health apps, there is a lack of tools specifically designed to cater to the unique needs of neurodivergent individuals, particularly those with ADHD and autism. Many existing apps do not allow for sufficient customization or fail to integrate features that help users understand their patterns and correlations between mood, symptoms, and external factors. Current solutions include mood tracking apps (e.g., Daylio, Moodfit, BIH etc.), symptom trackers, and general productivity tools. However, these often lack integration and advanced analytical capabilities that could provide insights tailored to neurodivergent users. There is a notable gap in applications that enable neurodivergent users to accessibly and comfortably track data and analyse it to provide actionable insights through pattern recognition and correlation spotting, especially within a specifically a neurodivergent user-friendly interface tailored for neurodivergent individuals.

PROJECT AIMS

The primary aim of this project is to design and develop a personalized digital assistant app that supports neurodivergent users in understanding and managing their moods, symptoms and/or medication, and daily activities. The assistant would serve as a comprehensive, modular and highly customizable assistive technology toolkit for individuals experiencing cognitive, executive function, and daily living challenges—including but not limited to those with neurodivergent conditions. The application aims to reduce barriers to independent living by providing adaptive, user-controlled solutions that minimize cognitive load while maximizing personalization capabilities. Through a unified platform with simple, intuitive controls, users will be able to tailor support tools to their unique needs across multiple life domains. The app will leverage techniques to identify patterns and provide insights that help users make informed decisions about their wellbeing. It will also signpost users to others with similar challenges and other supportive communities and ideally improve quality of life of users and utilize intelligent pattern recognition and predictive features positioned as aspirational enhancements for future development.

PROJECT OBJECTIVES

- **Conduct User Research:** Gather qualitative and quantitative data on the needs, preferences, and challenges faced by neurodivergent individuals using surveys, interviews, and focus groups.
- **Investigate:** Review existing assistive technology solutions, cognitive support applications, and accessibility frameworks to identify gaps in customizability, user control, and comprehensive support coverage.
- **Develop:** Ideate functionality modules addressing the ten primary challenge domains identified through user research:
 - Executive function and scheduling support
 - Communication and social navigation assistance
 - Learning and information processing aids
 - Workplace and educational accommodation tools
 - Sensory regulation and environmental control
 - Health, medication, and self-care management
 - Travel, navigation, and environmental adaptation
 - Physical motor and coordination support
 - Crisis management and emergency response
 - Financial management and independence tools
- **Define App Features:** Create a simplified control system that balances extensive customization options with low cognitive load, ensuring users can create highly bespoke configurations through intuitive interactions.
- **Design the User Interface:** Create wireframes and prototypes that prioritize usability and accessibility, ensuring the app is tailored to the needs of neurodivergent users. Incorporate accessibility-first design principles throughout the application, including alternative input methods, adjustable sensory presentation, and cognitive load reduction strategies.
- **Develop Backend Architecture:** Implement a backend system that updates the Frontend automatically and seamlessly

using RESTful principles, Firebase for data storage, and Dart and Flutter for broad compatibility.

- Test and Iterate: Conduct user testing sessions to gather feedback and iterate on both the app features and user interface.
- Documentation: Prepare comprehensive documentation for the app, including user manuals, technical specifications, and design rationale.

North star features

- Implement Machine Learning Algorithms: Utilize libraries such as scikit-learn and TensorFlow to develop and deploy models that can recognize patterns and correlations within user data.
 - signposting to appropriate resources
 - A user forum for sharing and co-creating solutions
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ALIGNMENT WITH PROGRAMME LEARNING OUTCOMES

This project aligns with the Computer Science programme learning outcomes by:

1. Software Engineering

- Applying software engineering principles to design and implement a complex, modular mobile application
 - Demonstrating proficiency in mobile development frameworks and user interface design
- The project, which involves the design and development of a customizable assistive mobile application for individuals managing cognitive and executive function challenges, demonstrates direct application of software engineering knowledge and skills. It reflects the ability to design, implement, and test complex systems in a systematic and principled manner. The inclusion of features such as symptom tracking, reminder systems, and adaptive user interfaces highlights a thorough understanding of software structure, functionality, and end-user needs.

The work also illustrates an understanding of the complete software development life cycle—spanning requirements gathering, design, implementation, testing, and evaluation—supported by the use of modern development environments and tools suited for mobile system deployment. Furthermore, the project embodies the principles of simplicity, robustness, and systematic organization by prioritizing clarity and accessibility in design. These principles are especially significant in developing applications that cater to users with diverse cognitive profiles, where intuitive interaction and fault-tolerant functionality are critical.

2. Computer Systems

- Utilizing database systems for efficient data storage, retrieval, and analysis
- The project aligns with the learning outcomes associated with computer systems through the integration of multiple technical components within a mobile computing environment. It demonstrates an understanding of core aspects of mobile architectures, including system frameworks, data synchronization, and device-specific operations such as notifications and offline storage management. The development process highlights awareness of how system components interact to ensure performance reliability across both iOS and Android environments.

The project also applies usability principles in practice. By focusing on customizability and adaptive user control, the project shows a keen awareness of system interaction design and accessibility. The incorporation of lightweight, resource-efficient systems ensures compatibility with varying device specifications while maintaining user-centred design standards.

3. Applications

- Establishing architectural foundations that support future integration of advanced technologies including machine learning and predictive analytics
- The project demonstrates strong alignment with application-based learning outcomes through the creation of a real-world software solution addressing issues related to cognitive load, accessibility, and self-management. Situated within the assistive technology domain, the application contributes to an expanding area of health and wellbeing support through mobile computing. It reflects the ability to apply technical knowledge to socially meaningful use cases, such as helping individuals organize and manage daily routines, appointments, and wellbeing data.

This work also reflects the ability to translate theoretical principles of usability and interaction design into functional, user-tested interface components. The final system prioritizes personalization, efficient navigation, and visual clarity—key elements in applied computing that promote inclusivity and ease of use.

4. Disciplinary Skills

- Evaluating system performance, usability, and effectiveness through empirical methods
- Documenting design decisions, implementation processes, and research findings

The student demonstrates well-developed disciplinary skills throughout the handling of the project lifecycle. This includes problem analysis, technical design, and iterative solution-building. The project addresses complex human–computer interaction challenges such as cognitive accessibility and task automation, showcasing a structured approach to solving practical software design problems.

Project management competencies are visible through effective planning, milestone tracking, and integration of iterative feedback cycles. The student also demonstrates an understanding of ethical and professional considerations through secure management of sensitive data and awareness of privacy implications associated with assistive technologies. These skills reflect the responsible conduct expected within the field of computer science.

5. Generic Knowledge and Transferable Skills

- Implementing accessibility standards and inclusive design practices
- Conducting user research to inform requirements gathering and iterative development

The project supports the development of transferable skills that extend beyond disciplinary knowledge. It demonstrates practical problem-solving ability through the formulation of novel design solutions responding to real-world cognitive accessibility needs. Collaboration and communication skills are evidenced by engagement with survey participants and the translation of user feedback into informed design changes.

Additionally, the documentation and reporting elements of the project showcase clear written communication of technical content. This ability to present analytical reasoning, design rationale, and results in an accessible format reflects competence in communicating complex information to diverse audiences, both technical and non-technical.

6. Attributes

The project cultivates core graduate attributes associated with interdisciplinary synthesis, independent learning, and reflective practice. It demonstrates the ability to connect theoretical concepts across software engineering, human–computer interaction, and assistive technology design, illustrating integrative understanding. The student applies critical thinking to evaluate user needs, prioritize features, and implement design trade-offs that balance flexibility with simplicity.

Furthermore, the project exemplifies intellectual independence and adaptability—the capacity to approach unfamiliar challenges through rigorous experimentation and evidence-based refinement. It also highlights effective communication of complex technical decisions through structured documentation and planned dissemination via the final report and presentation.

Methodology

Phase 1: Research and Requirements (Weeks 1-5)

September 20 - October 27 2025

- Conduct comprehensive literature review of existing research on digital tools for mental health, cognitive load theory, assistive technology research for neurodivergent people, and accessibility guidelines
- Engage with target users throughout the design and development process, employing methods such as, reviewing current solutions, surveys, and usability testing.
- Analyze existing applications across all ten challenge domains to identify strengths, weaknesses, and gaps
- Review questionnaire data to extract detailed user requirements and priority features
- Interview potential users to understand current coping strategies, pain points, and desired functionality

- Define detailed functional and non-functional requirements
- Create user personas representing diverse challenge profiles and use cases

Phase 2: Design and Architecture (Weeks 6-8)

October 28 – November 17, 2025

- Design modular application architecture supporting independent development and deployment of feature modules
- Create low-fidelity wireframes exploring navigation patterns and interface approaches
- Develop interaction design patterns that support both simplicity and deep customization
- Adopt an agile approach for the development of the app, allowing for iterative design, feedback, and adjustments based on user input.
- Design data models that support current functionality and future ML integration
- Create high-fidelity mock-ups incorporating accessibility requirements and visual design
- Develop configuration system architecture enabling template-based and advanced customization
- Plan database schema supporting efficient data storage, retrieval, and pattern analysis
- Establish RESTful API architecture for backend-frontend integration

Phase 3: Core Development (Weeks 9-18)

November 18, 2025 – January 26, 2026

- Implement foundational application framework and navigation structure
- Develop backend infrastructure with RESTful API endpoints
- Develop priority modules based on user research findings:
 - Executive function and scheduling support
 - Health and medication management
 - Reminder and notification system with advanced customization
 - Basic pattern visualization and data tracking
- Implement customization engine and configuration interface
- Develop data persistence layer and synchronization capabilities
- Create accessibility features including alternative input methods and sensory adjustments

Phase 4: Extended Development (Weeks 19-24)

January 27 – March 9, 2026

- Implement additional support modules across remaining challenge domains
- Develop resource integration system for educational materials and community connections
- Create pattern recognition and insight visualization features
- Implement data export and sharing capabilities
- Develop template system for saving and sharing configurations
- Regularly test the app with users to ensure it meets their needs and expectations, making adjustments to functionality and user interface based on continuous feedback.

Phase 5: Testing and Iteration (Weeks 25-28)

March 10 – April 6, 2026

- Conduct usability testing with individuals representing diverse cognitive profiles
- Perform accessibility audits using automated tools and manual evaluation
- Gather feedback on customization depth, interface simplicity, and feature utility
- Iterate on design and functionality based on user feedback
- Conduct performance testing and optimization
- Perform security review and privacy compliance verification

Phase 6: Documentation and Evaluation (Weeks 29-32)

- Complete comprehensive project documentation including architecture diagrams, design rationale, and implementation details
 - Evaluate application effectiveness through user satisfaction surveys and task completion metrics
 - Analyze collected usage data to identify successful features and areas for improvement
 - Document lessons learned and recommendations for future development
 - Prepare final project report and presentation materials
 - Identify specific pathways for future ML integration based on collected data and user feedback
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PROJECT MILESTONES

Milestone 1 (Week 5): Requirements specification document, literature review summary, user research findings, and initial design concepts

Due: October 21, 2025

Milestone 2 (Week 8): Complete application architecture design, high-fidelity mock-ups, data model specifications, and technical design document

Due: November 11, 2025

Milestone 3 (Week 12): Working prototype with core modules (executive function support, medication management, basic tracking), functional navigation, and customization framework

Due: December 9, 2025

Milestone 4 (Week 18): Feature-complete core application with all priority modules implemented, advanced customization capabilities, and insight visualization

Due: January 20, 2026

Milestone 5 (Week 24): Extended application with all planned modules, resource integration, accessibility features, and template system

Due: March 3, 2026

Milestone 6 (Week 28): Tested and refined application incorporating user feedback, performance optimization, and accessibility compliance

Due: March 31, 2026

Milestone 7 (Week 32): Final deliverables including completed application, comprehensive documentation, evaluation report, and project presentation

Due: April 28, 2026

REQUIRED KNOWLEDGE/SKILLS/TOOLS/RESOURCES

Existing Knowledge and Skills

- Application development (iOS/Android platforms)
- User interface and user experience design principles
- Database design and management
- Software engineering methodologies and design patterns
- Version control and collaborative development tools
- Familiarity with Python programming, including libraries such as scikit-learn, TensorFlow, and Flask/FastAPI.
- API integration and data synchronization

Knowledge and Skills to Develop

- Knowledge/Skills:
Flutter + dart familiarity

- Understanding of Firebase and its integration with mobile applications.
- Knowledge of pattern recognition implementations and data analysis.
- Privacy and security best practices for health-related applications
- Architectural patterns supporting future ML integration
- Accessibility standards (WCAG guidelines) and implementation techniques
- Cognitive load theory and its application to interface design
- Assistive technology frameworks and platform-specific accessibility APIs
- Advanced customization and configuration system design
- Data visualization for non-technical audiences

Tools and Technologies

Development Environment:

- vs code, android studio
- Version control and project management: GitHub
- - Database management system (Firebase)
- Design Tools: Figma or Adobe XD
- Testing Tools: testing libraries, surveys or direct feedback sessions with target users.

Accessibility Tools:

- Platform screen readers (VoiceOver, TalkBack)
- Accessibility testing frameworks
- Color contrast analyzers
- Alternative input testing devices

Testing and Evaluation:

- Usability testing platforms
- Analytics tools (privacy-preserving)
- Performance monitoring tools
- Automated testing frameworks

Research Resources:

- Academic databases (ACM Digital Library, IEEE Xplore, Google Scholar)
- Accessibility guidelines documentation
- Assistive technology research literature
- User testing participants representing diverse cognitive profiles

Hardware Resources

- Development computer (Mac for iOS, or appropriate system for Android/cross-platform)
- Testing devices representing range of screen sizes and capabilities
- Alternative input devices for accessibility testing
- Optional: Wearable devices for future integration exploration

Timeplan

Assistive Technology Mobile App Development Timeline

